CourseCompass: Association Rule Mining–Based Course Recommendation System

Applied Machine Learning (AI3002) — Project Assignment #01 Technical Report

Institution: National University of Computer & Emerging Sciences (FAST-NUCES)
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1. Introduction and Problem Statement

In modern online education, learners face vast catalogs without individualized guidance. Unlike black-box collaborative filtering, **Association Rule Mining (ARM)** provides fully transparent, interpretable, and mathematically verifiable recommendations. CourseCompass formulates each learner's enrollments as a transaction basket, discovers frequent course co-occurrences via the Apriori algorithm, calculates direction-sensitive metrics (Support, Confidence, Lift, Leverage, Conviction), strictly excludes already-enrolled items, and presents an interactive Streamlit application with 'Why this recommendation?' rationales.

2. Application Domain and Motivation

Domain: Academic Course Enrollment Analytics on edX/HarvardX MOOCs. Open online learning suffers from massive dropout rates (>90%), largely due to isolated course taking. CourseCompass identifies natural interdisciplinary bridges—such as connecting Classical Humanities to Moral Philosophy, or Public Health Biostatistics to Computer Science—guiding students along coherent educational pathways.

3. Dataset Description & Preprocessing Pipeline

The system uses the official de-identified **HarvardX Person-Course Academic Year 2013 Dataset** (Harvard Dataverse DOI: 10.7910/DVN/26147). The dataset contains **338,223 total records** across **301,609 unique learners** and 5 flagship HarvardX courses.

Dataset Metric	Value	Description
Total Records	338,223	Total student course registrations
Unique Learners	301,609	Distinct de-identified user accounts
Unique Courses	5	Flagship HarvardX inaugural courses
Multi-Course Baskets	30,267 (10.0%)	Learners with basket size ≥ 2 (Primary ARM cohort)
Single-Course Baskets	271,342 (90.0%)	Learners taking exactly one course
Max Basket Size	5	Enrolled in all 5 flagship courses

4. Association Rule Mining Methodology

Mathematical Formulations:
• **Support:** $\text{support}(A \to B) = P(A \cap B) = \text{count}(A \cup B) / N$
• **Confidence:** $\text{confidence}(A \to B) = P(B \mid A) = \text{support}(A \cup B) / \text{support}(A)$
• **Lift:** $\text{lift}(A \to B) = \text{confidence}(A \to B) / \text{support}(B)$
• **Leverage:** $\text{leverage}(A \to B) = \text{support}(A \cup B) - \text{support}(A) \cdot \text{support}(B)$
• **Conviction:** $\text{conviction}(A \to B) = (1 - \text{support}(B)) / (1 - \text{confidence}(A \to B))$

Mathematical Derivation: Why Confidence(A → B) ≠ Confidence(B → A):
While Joint Support $P(A \cap B)$ is symmetric, Confidence divides by the antecedent's individual support: $\text{conf}(A \to B) = P(A \cap B)/P(A)$ while $\text{conf}(B \to A) = P(A \cap B)/P(B)$. Because course popularities vary, $P(A) \neq P(B)$. Empirically, conf(Greek Hero → Justice) = 26.09% / 39.02% = **66.87%**, whereas conf(Justice → Greek Hero) = 26.09% / 57.85% = **45.11%**. The 21.76% difference reflects that Justice is broader and more populous than Greek Hero.

5. Apriori Explanation and Hand-Worked Example

The Apriori Principle (Downward Closure): If an itemset is frequent, all its subsets must be frequent. Any candidate itemset containing an infrequent subset is pruned prior to database scanning.

Hand-Worked Example (N=5 transactions, min_sup=0.40):

Baskets: T1={CS, Ethics}, T2={CS, Stats, Ethics}, T3={Stats, Bio}, T4={CS, Stats, Bio}, T5={CS, Ethics}.

• Level 1: C1={CS:4, Ethics:3, Stats:3, Bio:2} $\rightarrow$ All survive into L1.

• Level 2: C2 joined from L1. {CS, Bio} has count 1 (pruned), {Ethics, Stats} has count 1 (pruned). L2={{CS, Ethics}, {CS, Stats}, {Stats, Bio}}.

• Level 3: Joining {CS, Ethics} and {CS, Stats} produces {CS, Ethics, Stats}. Its subset {Ethics, Stats} $\notin$ L2. **Pruned by Apriori property without scanning!**

6. Frequent Itemsets and Association Rules

Mining on multi-course learner baskets (N=30,267) at min_sup=0.05 and min_conf=0.15 yields 15 itemsets and 25 rules:

Discovered Association Rule (A → B)	Sup	Conf	Lift	Leverage
{CB22x: Greek Hero} → {ER22x: Justice}	0.2609	66.87%	1.16x	+0.0352
{ER22x: Justice} → {CB22x: Greek Hero}	0.2609	45.11%	1.16x	+0.0352
{PH207x: Health Numbers} → {CS50x: Intro to CS}	0.1597	54.70%	1.10x	+0.0150
{PH278x: Environmental} → {ER22x: Justice}	0.2426	53.50%	0.92x	-0.0197
{CB22x: Greek Hero} → {CS50x: Intro to CS}	0.1361	34.87%	0.70x	-0.0574
{CB22x, CS50x} → {ER22x: Justice}	0.0616	45.24%	0.78x	-0.0171

7. Recommendation Engine Logic

Recommendation Process:

1. Matches all association rules whose antecedent is a subset of the learner's basket.
2. **Strict Exclusion:** Completely eliminates courses already selected by the learner.
3. **Evidence Aggregation:** Collects all rules supporting each candidate course.
4. **Ranking:** Supports Confidence-First, Lift-First, and Hybrid Score (Conf × √Lift).
5. **Dynamic Explanation:** Produces natural-language 'Why this recommendation?' narratives.

8. Threshold Experiments & Outcome Comparison

Experiment Dimension	Setting A (Conservative)	Setting B (Exploratory)
Minimum Support (min_sup)	0.10 (10%)	0.03 (3%)
Minimum Confidence (min_conf)	0.40 (40%)	0.15 (15%)
Frequent Itemsets Discovered	13 itemsets	16 itemsets
Association Rules Generated	9 rules	26 rules
Mean Rule Confidence	52.6%	38.4%
Mean Rule Lift	1.01x	0.86x
Recommendation Behavior	High precision, low coverage	Broad coverage, more serendipity

9. Manual Rule Evaluation (10 Discovered Rules)

Rule	Conf / Lift	Category	Domain Rationale
{Greek Hero} → {Justice}	66.9% / 1.16x	**Useful**	Bridges classical literature with moral philosophy.
{Justice} → {Greek Hero}	45.1% / 1.16x	**Useful**	Reciprocal humanities bridge for ethics students.
{Health Numbers} → {CS50x}	54.7% / 1.10x	**Useful**	Bridges quantitative biostatistics to computer programming.
{Environmental} → {Justice}	53.5% / 0.92x	**Useful**	Connects climate and environmental health to social justice.
{Justice} → {CS50x}	37.6% / 0.76x	**Obvious**	CS50x is taken by 50% of students; driven by platform size.
{Greek Hero} → {CS50x}	34.9% / 0.70x	**Obvious**	Reflects baseline global popularity rather than domain affinity.
{Health Numbers} → {Environmental}	40.8% / 0.90x	**Useful**	Domain coherence within Harvard Public Health curriculum.
{Environmental} → {Health Numbers}	26.3% / 0.90x	**Weak**	Low confidence; environmental students don't reliably take biostats.
{Greek Hero, CS50x} → {Justice}	45.2% / 0.78x	**Unexpected**	Students combining coding and Greek hero converge on justice.
{Greek Hero, Justice} → {CS50x}	23.6% / 0.48x	**Weak**	Low confidence and poor lift; pure humanities rarely cross to CS.

10. System Limitations

1. **Lack of Pedagogical Sequence:** ARM treats baskets as unordered sets. It cannot capture prerequisite order (e.g., taking CS50x before Advanced Data Science vs. after).
2. **Popularity Bias:** Mega-courses like CS50x dominate association rules due to sheer scale (50% enrollments). Lift and Leverage must be strictly checked to avoid recommending popular courses indiscriminately.
3. **Cold-Start Problem:** Brand-new courses with zero enrollment history have 0 support and cannot be recommended by Apriori.